

Data Preparation Report

Week 1 – International Student Compliance & Eligibility Data

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Scope: SEVIS extract and Connect (SLU Connect) extract, Saint Louis University, April 2026 export

A note on data scope: the brief references three source systems – Applicant, Connect, and SEVIS. Only Connect and SEVIS extracts (plus a reduced-column copy of Connect) were available for this submission; no Applicant dataset was provided. This is documented as a key limitation in Section 3, and the join/integration analysis below is based on what could be established from the two available extracts.

1. Dataset Overview & Documentation

1.1 Dataset Inventory

Dataset	File	Rows	Columns	Grain
SEVIS	SEVIS_Data_-_Sevisapril2026.csv	3,524	126	One row per non-immigrant student record (SEVIS_ID)
Connect	Connect_Data_-_Connectapril2026__1_.csv	34,341	24 (8 populated, 16 fully blank)	One row per case-review touchpoint (Reference_ID repeats)
Connect (reduced copy)	Copy_of_Connect.xlsx (sheet: SLUConnect)	34,341	8	Same underlying export as Connect CSV, limited to the populated columns

Applicant dataset: not supplied. The report structure below flags every point where this file would normally be required.

1.2 Variable Types

SEVIS (126 columns) breaks down approximately as follows:

Category	Example fields	Approx. count
Identifiers (ID)	SEVIS_ID, NonImmigrant_ID, FIN_ID, School_Code, Campus_Code, Major_1_Code	~10
Categorical	SEVIS_Status, Class_of_Admission, Gender, Education_Level, I_20_Issue_Reason, Country_of_Citizenship	~35
Date	Date_of_Birth, Status_Change_Date, Program_Start_Date, Program_End_Date, Visa_Issuance_Date, I_901_Transaction_Date	~20
Numerical (financial / counts)	Tuition_Fees, Living_Expenses, Student_Personal_Funds, I_901_Transaction_Amount, Number_of_Months_(for_funding)	~15
Boolean / flag (Y/N)	CPT_Approved, STEM_OPT_Approved,	~25

	Student_Has_English_Proficiency, Cap_Gap_Extended, Student_Transfer_Out	
Free text / address / comment	US_Physical_Address_1, Foreign_Address_1, Overall_Remarks, Other_Cost_Comment	~20

Connect (24 columns, 8 populated) breaks down as:

Category	Fields	Notes
Identifier	Reference_ID	Case-level key; repeats across review touchpoints (see 1.3)
Categorical / status	Deposit_Status, I-20_Status, University	Yes/No flags plus a constant institution field
Free text (staff names)	Assigned, Recieved_At	Comma-separated advisor names; Recieved_At also stores a timestamp string
Date/time	Created_At, Modified_At	Unix epoch milliseconds
Fully blank in this extract	Bank_Statement, Sponsor_Letter, Passport, Total_Funding_Amt, Cost_To_Issue_I-20, Main_Status, Date_of_Review, Addtl_Financial, Tuition_and_Fees, Est_books_supplies_insurance, Est_Living_Expenses, Total_Scholarship, Deposit_Date, English_Language_Requirement, ELC_Reply_Form_Status, ELC_Summer_Student_Response	16 columns, 100% null – the schema exists but was not populated for this export cycle

1.3 Key Identifiers

- SEVIS_ID – unique per row in the SEVIS extract (3,524 unique values across 3,524 rows). This is the natural primary key for the SEVIS table.
- Reference_ID – present in Connect but not unique per row: 34,341 rows resolve to only 6,875 distinct Reference_ID values (average ~5.0 rows per case; 6,855 of the 6,875 cases have exactly 5 rows). Reference_ID is therefore a case-level key, and Connect is best understood as a review/touchpoint log rather than one-row-per-applicant table.
- No shared key currently links SEVIS to Connect. There is no common ID, email, or name field between the two extracts – University is present in both but is constant ("Saint Louis University") in every row of both files, so it cannot be used to join records. This is expanded on in Section 3.

1.4 General Dataset Themes

Theme	Where it shows up
Eligibility	SEVIS: SEVIS_Status, Class_of_Admission, Education_Level, I_20_Issue_Reason, Program_Start/End_Date
Document / compliance	Connect: Bank_Statement, Sponsor_Letter, Passport, Deposit_Status, I-20_Status

status	(schema present, values blank in this extract); SEVIS: Passport_Number, Visa_Number, Student_Has_English_Proficiency
Government fee payment	SEVIS: I_901_Transaction_Type, I_901_Transaction_Date, I_901_Transaction_Amount, I_901_Fee_Payment_Receipt (the SEVIS I-901 fee)

1.5 Initial Structural Observations

- Both extracts represent a single institution (Saint Louis University) and, in SEVIS, a single admission class (F-1) and status (INITIAL) – these three fields carry zero variance and add no discriminating value in their current form.
- SEVIS is wide (126 columns) but many columns are fully or near-fully null (e.g., SEVIS_Legacy_* names, Thesis_Dissertation, Study_Research_Abroad, Agency_Code are 100% blank; FIN_ID, Passport_Number, Visa_Number, US_Telephone are 90–99% blank).
- Connect is long and narrow in practice: only 8 of 24 columns carry any data, and the table is structured as repeated case-review events rather than one row per applicant.
- The xlsx copy of Connect matches the CSV in row count (34,341) and appears to be the same source table pre-filtered to its 8 populated columns; a naming inconsistency exists between the two (I-20_Status in the CSV vs I_20_Status in the xlsx), which is noted for standardisation.

2. Data Cleaning & Processing Summary

2.1 Missing Value Analysis

Missing values are recorded as the literal string "NULL" in both source files rather than a true empty/NaN value; this was standardised first so completeness could be measured. Selected columns of interest:

Dataset	Column	Missing	% Missing
SEVIS	FIN_ID	3,461	98.2%
SEVIS	Passport_Number	3,212	91.1%
SEVIS	Visa_Number	3,212	91.1%
SEVIS	I_901_Transaction_Type / Amount / Receipt	1,869	53.0%
SEVIS	SEVIS_ID / SEVIS_Status / Email / Country_of_Citizenship / Program dates	0	0.0%
Connect	16 schema columns (Bank_Statement, Passport, Main_Status, financial fields, etc.)	34,341	100%
Connect	Assigned	19,810	57.7%
Connect	Reference_ID / Deposit_Status / University / Recieved_At / I-20_Status	0	0.0%

The 16 fully-null Connect columns are flagged for the data owner: either the fields are genuinely not yet in use for this cohort, or the export excluded populated source columns. Until clarified, these columns should not be relied on for compliance analysis (e.g., document-verification status cannot currently be assessed from Connect – see Section 4).

2.2 Duplicate Record Checks

- Exact full-row duplicates: 0 in SEVIS, 0 in Connect.
- SEVIS_ID uniqueness: confirmed unique across all 3,524 SEVIS rows – no duplicate student records.
- Reference_ID repetition in Connect (34,341 rows → 6,875 cases) was checked to confirm these are not accidental duplicates: rows sharing a Reference_ID differ in Assigned/Recieved_At (i.e., different advisor touchpoints) even when Deposit_Status and I-20_Status stay constant for that case, confirming this is a genuine event log rather than duplicated records.

2.3 Standardisation of Categories & Formats

Issue	Detail	Treatment
Literal "NULL" strings	Both files use the text "NULL" instead of a true null	Converted to proper NULL/NaN on load
Date fields stored as text	e.g., Date_of_Birth, Program_Start_Date in MM/DD/YYYY string format	Cast to DATE type for PostgreSQL load
Epoch timestamps	Connect Created_At/Modified_At in Unix epoch milliseconds	Converted to timestamp type
Column-name inconsistency	I-20_Status (Connect CSV) vs I_20_Status (Connect xlsx)	Standardised to a single column name (I_20_Status) before integration
Mixed data type in one field	On_Campus_Employment mixes numeric values (0, 1000) with a date-like artifact value "12/30/1899" (an Excel epoch-zero bug) across 3,524 rows	Flagged for correction; "12/30/1899" recoded as 0/unknown rather than treated as a real date
Case/format of categorical fields	Education_Level, Class_of_Admission, SEVIS_Status already consistently upper-cased in source	No change needed; confirmed consistent
Multi-value text fields	Connect Assigned/Recieved_At store comma-separated staff names in a single cell	Left as-is for Week 1 (not required for eligibility/compliance fields); flagged for later parsing if advisor-workload analysis is needed

2.4 Preprocessing Steps Performed Prior to Integration

- Standardised missing-value representation ("NULL" → NaN) across both extracts.
- Parsed all date columns from MM/DD/YYYY text to proper DATE types; converted Connect epoch timestamps to TIMESTAMP.
- Dropped/quarantined the 16 fully-null Connect columns from the Week 1 working table (kept in the raw staging table, excluded from the cleaned table) pending confirmation from the data owner.
- Reconciled the I-20 status column name between the two Connect exports.
- Verified primary-key uniqueness (SEVIS_ID) and documented the case-level key behaviour of Reference_ID before attempting any joins.

3. Data Integration & Validation

3.1 Planned Join Logic

The intended integration path for this project is: Applicant ↔ Connect (via a shared applicant/case ID) and Applicant ↔ SEVIS (via a shared student ID), with Applicant acting as the bridge table that resolves both systems to a single student-level record.

Applicant dataset not received: with only Connect and SEVIS available, no shared key could be identified between them directly (see 3.2). The join logic above is therefore documented as the target design, not yet executed.

3.2 Fields Checked for a Direct SEVIS ↔ Connect Relationship

Candidate field	Present in SEVIS	Present in Connect	Usable as join key?
Student ID / SEVIS_ID / Reference_ID	Yes (SEVIS_ID)	Yes (Reference_ID)	No – different ID systems, no crosswalk available
Name	Yes (Surname/Given_Name)	No (Connect stores advisor names, not student names)	No
Email	Yes	No	No
University	Yes (constant: Saint Louis University)	Yes (constant: Saint Louis University)	No – zero variance, matches every row to every row
Date fields (e.g., created/modified vs program start)	Yes	Yes	No – timestamps reflect different processes (case handling vs. SEVIS status events) and do not align 1:1

Conclusion: no reliable join key currently exists between the SEVIS and Connect extracts. Any attempt to join on row order or University would produce a Cartesian-style mismatch (3,524 SEVIS rows against 34,341 Connect rows / 6,875 Connect cases) and cannot be validated.

3.3 Before vs After Row Counts

Step	SEVIS rows	Connect rows (raw)	Connect cases (distinct Reference_ID)
Raw import	3,524	34,341	6,875
After NULL-string standardisation	3,524	34,341	6,875
After dropping fully-blank Connect columns (columns only, no rows removed)	3,524	34,341	6,875
After integration with SEVIS	Not performed – no valid key	Not performed	Not performed

3.4 Record Reconciliation Checks Performed

- Row-count integrity: no rows were lost or duplicated during type conversion or NULL standardisation for either extract (row counts identical before/after cleaning).
- Key uniqueness: SEVIS_ID confirmed unique (100%); Reference_ID confirmed non-unique with a consistent, explainable pattern (~5 touchpoint rows per case), not a data-quality error.
- Cross-file consistency: confirmed the Connect CSV and Connect.xlsx represent the same 34,341-row export (same Reference_ID population, same value patterns), differing only in which columns were retained.

3.5 Identified Inconsistencies Across Systems

- No common identifier between SEVIS and Connect (Section 3.2).
- Column-naming drift between two exports of the same Connect table (I_20_Status vs I_20_Status).
- 16 of 24 Connect columns – including every document-compliance field (Bank_Statement, Sponsor_Letter, Passport) and every financial field – are entirely blank in this extract, despite the schema suggesting they should hold data.

3.6 Data Limitations & Risks for Later Analysis

- Missing Applicant dataset is the single biggest risk: without it, SEVIS and Connect cannot be integrated at the student level, and any Week 2 compliance analysis that assumes a combined view will need this file first.
- Document-compliance fields in Connect (bank statement, sponsor letter, passport) contain no data in this extract, so document-submission completeness cannot yet be measured from Connect alone (Section 4).
- SEVIS financial-support fields (Passport_Number, Visa_Number, FIN_ID) are 90–98% missing, which may reflect either genuine data gaps or fields that are simply not required at the INITIAL stage – this needs a subject-matter check before being treated as a compliance gap.
- The On_Campus_Employment mixed-type values (numeric vs. "12/30/1899") should be corrected at the source, or explicitly recoded, before any numerical analysis of that field.

4. Initial Structural Observations

This section reports what the data shows structurally; deeper exploratory analysis is planned for Week 2 once the Applicant dataset is available.

4.1 Eligibility Distribution (SEVIS)

Field	Distribution
Education_Level	MASTER'S 3,064 (86.9%) · BACHELOR'S 357 (10.1%) · OTHER 45 · DOCTORATE 36 · LANGUAGE TRAINING 22
Class_of_Admission / SEVIS_Status / I_20_Issue_Reason	100% F-1 / 100% INITIAL / 99.5% "INITIAL" issue reason – this cohort is almost entirely new F-1 initial-status students, with a small number of change-of-status (9), transfer (7), and change-of-education-level (2) cases
Top countries of citizenship	India 2,089 (59.3%) · Ghana 326 · Nigeria 302 · Pakistan 117 · Zimbabwe 106 · Nepal

	90 · Kenya 89 · Bangladesh 62
Top majors	Computer & Information Sciences, General (927) · Computer Science (676) · Informatics (423) · Medical Informatics (269) · Cyber/Electronic Operations & Warfare (141)

4.2 Document Submission Completeness (Connect)

The three document-status fields intended to track compliance – Bank_Statement, Sponsor_Letter, and Passport – are 100% blank across all 34,341 Connect rows in this extract. Deposit_Status and I-20_Status are fully populated (Yes/No), giving a partial picture:

Field	Yes	No
Deposit_Status	7,526 (21.9%)	26,815 (78.1%)
I-20_Status	11,911 (34.7%)	22,430 (65.3%)

This is a structural gap, not an analytical finding: document-verification completeness cannot be assessed until the Bank_Statement/Sponsor_Letter/Passport fields are populated in a future export.

4.3 Government Fee Payment Coverage (SEVIS I-901)

I_901_Transaction_Type	Count	% of SEVIS rows
Payment	1,602	45.5%
Cancellation	53	1.5%
Not recorded (blank)	1,869	53.0%

Where a payment is recorded, the amount takes one of three fixed values (350 being the most common), consistent with standard SEVIS I-901 fee tiers. Just over half of records show no I-901 transaction at all, which is worth confirming with the data owner: it may indicate fee payment tracked elsewhere, exemption, or a genuine compliance gap.

4.4 Compliance-Related Data Gaps Identified

- No Applicant-level bridge between SEVIS and Connect (Section 3).
- Connect's document-verification fields are unpopulated in this extract (4.2).
- 53.0% of SEVIS records show no I-901 fee transaction data (4.3).
- Passport_Number and Visa_Number are missing for over 90% of SEVIS records, limiting identity/travel-document cross-checks.
- On_Campus_Employment contains a mixed-type artifact value that should be resolved before it is used in any compliance rule.

Summary & Next Steps

- Request the Applicant dataset (or the crosswalk table linking Connect Reference_ID to SEVIS_ID) before Week 2 EDA – this is the primary blocker to a fully integrated dataset.

- Confirm with the data owner whether the 16 blank Connect columns (including all document-compliance fields) are expected to be empty for this export or should be re-pulled.
- Load the cleaned SEVIS and Connect tables into PostgreSQL as separate staging tables now; integration will be a follow-up step once a valid key is confirmed.
- Resolve the On_Campus_Employment mixed-type field and the I-20 status column-naming inconsistency at the source before Week 2.